



# Hands-On Lab Exercise 5: Conditional Statements in Shell Script

**Topic:** Decision Making Using Conditional Statements ( `if` )

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This lab explains how shell scripts **make decisions** using conditional statements.

Conditional statements allow scripts to: - Think logically - Compare values - Take decisions - Choose actions - Perform smart automation

Without conditions, scripts only execute commands. With conditions, scripts become **intelligent automation systems**.

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## What are Conditional Statements?

Conditional statements allow the script to:

Perform actions based on conditions (true or false)

In simple words:

"If something is true → do this, otherwise → do something else"

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## Why Do We Need Conditions?

We need conditions because: - Automation must make decisions - Systems behave differently - Inputs change - Environments change - Logic is required - Real automation needs intelligence

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## Purpose of Conditional Statements

Conditions help scripts to: - Control flow - Validate inputs - Check system state - Make decisions - Perform smart automation - Build real systems

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## Learning Objective

- Understand condition concept
  - Learn `if` syntax
  - Compare numbers
  - Build logical scripts
  - Create decision-based automation
- 



## Learning Outcome

After this lab, participants will: - Use `if` statements - Compare values - Write logical scripts - Build intelligent automation - Create condition-based programs

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## Basic `if` Syntax in Shell

```
if [ condition ]  
then  
    commands  
fi
```

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## Example 1: Number Comparison Script



### Purpose:

Compare two numbers and show which is greater.

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### Step 1: Create Script

```
touch compare.sh
```

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## Step 2: Open File

```
nano compare.sh
```

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## Step 3: Write Script

```
#!/bin/bash

echo "Enter first number:"
read num1

echo "Enter second number:"
read num2

if [ $num1 -gt $num2 ]
then
    echo "$num1 is greater than $num2"
fi

if [ $num1 -lt $num2 ]
then
    echo "$num2 is greater than $num1"
fi

if [ $num1 -eq $num2 ]
then
    echo "Both numbers are equal"
fi
```

---

## Step 4: Permission

```
chmod +x compare.sh
```

---

## Step 5: Run

```
./compare.sh
```



## Example 2: Prime Number Check Script



### Purpose:

Check whether a number is prime or not.

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### Step 1: Create Script

```
touch prime.sh
```

---

### Step 2: Open File

```
nano prime.sh
```

---

### Step 3: Write Script

```
#!/bin/bash

echo "Enter a number:"
read num

count=0

i=1
while [ $i -le $num ]
do
    if [ $(( num % i )) -eq 0 ]
    then
        count=$(( count + 1 ))
    fi
    i=$(( i + 1 ))
done

if [ $count -eq 2 ]
then
    echo "$num is a Prime Number"
else
```

```
    echo "$num is NOT a Prime Number"  
fi
```

---

## Step 4: Permission

```
chmod +x prime.sh
```

---

## Step 5: Run

```
./prime.sh
```



## Example 3: Leap Year Script



### Purpose:

Check whether a year is a leap year or not.

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## Step 1: Create Script

```
touch leap.sh
```

---

## Step 2: Open File

```
nano leap.sh
```

---

## Step 3: Write Script

```
#!/bin/bash  
  
echo "Enter year:"
```

```
read year

if [ $(( year % 400 )) -eq 0 ]
then
    echo "$year is a Leap Year"

elif [ $(( year % 100 )) -eq 0 ]
then
    echo "$year is NOT a Leap Year"

elif [ $(( year % 4 )) -eq 0 ]
then
    echo "$year is a Leap Year"

else
    echo "$year is NOT a Leap Year"
fi
```

---

## Step 4: Permission

```
chmod +x leap.sh
```

---

## Step 5: Run

```
./leap.sh
```

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## Example 4: OS Detection and Package Installation Script

### Purpose:

Automatically detect OS and install package accordingly.

Scenario: Some servers use Red Hat, some use CentOS.  
Script must detect OS and install package.

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## Step 1: Create Script

```
touch install_os.sh
```

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## Step 2: Open File

```
nano install_os.sh
```

---

## Step 3: Write Script

```
#!/bin/bash

os=$(cat /etc/os-release | grep ^ID= | cut -d= -f2)

echo "Detected OS: $os"

if [ "$os" = "rhel" ]
then
    echo "Red Hat detected"
    yum install -y git
    echo "Git installed on Red Hat"
fi

if [ "$os" = "centos" ]
then
    echo "CentOS detected"
    yum install -y git
    echo "Git installed on CentOS"
fi

if [ "$os" = "ubuntu" ]
then
    echo "Ubuntu detected"
    apt update
    apt install -y git
    echo "Git installed on Ubuntu"
fi
```

---

## Step 4: Permission

```
chmod +x install_os.sh
```

---

## Step 5: Run

```
./install_os.sh
```

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## Understanding the Logic

- Script checks conditions
  - Makes decisions
  - Executes correct command
  - Automates smart actions
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## Key Understanding

- `if` enables decision making
  - Conditions create intelligence
  - Scripts become smart systems
  - Automation becomes logical
- 



## Summary

- Conditional statements = decision making
  - `if` is core automation logic
  - Scripts can think logically
  - Scripts adapt to environment
  - Real automation needs conditions
- 



## End of Lab

This lab transforms shell scripts into **intelligent automation systems**.



Now scripts can: - Decide - Compare - Analyze - Act smartly

This is the foundation of: - DevOps automation - Cloud automation - CI/CD pipelines - Infrastructure automation

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